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Developing Academic Talent: DGMT-based principles and best practices[1]

Here are very common slogans about talent development: "Optimize your talents!" "Develop your capacities to their utmost!" "Achieve your full potential!" These slogans, or similar ones, appear everywhere: in the media, on billboards, in motivational documents, literally everywhere. Put any of them in a Google search, and it will propose thousands of web pages. More importantly for the present subject, politicians and school administrators around the world have chosen this dream as the fundamental goal for their national or regional educational systems. In doing so, they were following the lead given by the United Nations Organization in its Convention on the Rights of the Child, signed in 1989. Its Article 29 states: "the education of the child shall be directed to (…) the development of the child's personality, talents and mental and physical abilities to their fullest potential" (United Nations, 1989).

That basic goal statement, reaching one's full potential, expresses two basic assumptions. First, it assumes the existence of human potentialities that are distinct from human achievements. Second, it assumes that these two concepts are related: achievements have their origin in human potentialities. We give these potentialities different labels: aptitudes, capacities, innate abilities, innate talents, or natural abilities. I will use here the label 'aptitude,' because I believe it conveys best what people have in mind when they think about potentialities. Achievements also have many labels: accomplishment, competence, expertise, realization, or systematically developed abilities. Aptitudes are distinct from achievements in many ways, but especially because of their spontaneous and easy development, as opposed to the efforts people deploy to acquire competencies. That distinction between aptitudes and achievements, especially between their high-level manifestations as giftedness and talent, constitutes the fundamental premise of the theoretical framework I proposed more than three decades ago to explain academic talent development, and which I have continued to develop since (Gagné, 1985, 2004, 2020). That theoretical framework is named *Differentiating Model of Giftedness and talent*, or DMGT (in German, Differenzierendes Modell von Begabung und Talent).

In the first part of this chapter, I will briefly survey the DMGT. In the second part, I will demonstrate that most of us greatly underestimate the size of individual differences in academic achievement, especially among talented students; I will also argue that school systems around the world ignore these individual dif-

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ferences, as well as the basic educational needs of talented students. I will finally describe what 'real' academic talent development (ATD) should look like.

1. A theoretical framework for ATD

Let us begin this overview with the two key concepts of competence and aptitude.

1.1 Competence

Competence, the outcome of any learning process, manifests itself in two main types: factual knowledge, and mental or physical skills. Each human occupation is defined by the mastery of a specific set of competencies; diplomas usually confirm that mastery. In the universe of work there are thousands of occupations; they are represented in the DMGT (see Figure 1)[2] through nine major fields: academic, technical, scientific, artistic, social service, sales and marketing, business operations, games, and sports. Each occupation possesses its competency package; it is the curriculum that students need to learn if they want to access that occupation. But the concept of competence applies long before students reach the level of occupational training. Youngsters must first master academic competencies: the knowledge and skills built into the primary and secondary school curriculum.

Academic competence varies qualitatively, according to the different subject matters; it also varies quantitatively, from insufficient mastery, which will lead to repeating a grade or being refused a diploma, to outstanding mastery, which will place the students among the top achievers in their learning group. People commonly use the terms excellence or expertise as label for that high mastery; I have chosen to use the term talent. Academically talented students can excel in just one subject matter, or in most of them. Of course, the term talent applies as well to all occupational fields covered by the nine categories: talented engineers, talented mechanics, talented musicians, talented nurses, talented teachers, etc.

If talented means an 'outstanding' level of competence, how competent do students, workers, or professionals need to be to deserve that label? Should they be among the top 20 %, the top 10 %, or the top 1 %? My own choice is top 10 %, and you will find in my published work (Gagné, 1998) a detailed discussion of that particular choice of percentage to establish the prevalence of talented individuals, in other words the size of the talented population. But there are levels of excellence within that population; for instance, players in a National Football League demonstrate much more talent than those who just excel in a local league. Consequently, the DMGT proposes four levels of talent (formerly five) based on the metric system (see Figure 2). It starts with the basic threshold of top 10 %; then

2 All figures are printed at the end of this article.

we have the highly talented (top 1 %), the exceptionally talented (top 1:1,000), and finally the extremely (or profoundly) talented (top 1:10,000).

1.2 Aptitudes

General notions. Aptitudes are present at the beginning of every learning process, and serve as the building blocks for the competencies to be mastered. Aptitudes also differ qualitatively and quantitatively. The DMGT proposes six qualitatively distinct aptitude domains (see Figure 1): intellectual aptitudes (e.g., abstract reasoning, memory, spatial ability), creative aptitudes (e.g., invention, problem solving, visual or musical creativity), social aptitudes (e.g., leadership, social perspicacity, salesmanship), perceptual aptitudes (e.g., visual or smell acuity, kinaesthesia), muscular aptitudes (e.g., strength, speed, endurance), and motor control aptitudes (e.g., reflexes, dexterity, equilibrium). Research has shown that all aptitudes have important biological and hereditary foundations. Researchers have used all kinds of comparisons to identify the relative influences of Nature and Nurture; they compared for instance the similarity between identical and fraternal twins, between adopted siblings raised in the same family, between parents and their blood or adopted children. All these studies amount to one key conclusion: whatever the aptitude, genetics plays a substantial role in the explanation of individual differences, a role approximately equivalent to that of environmental influences.

Within each domain, aptitudes vary quantitatively, from minimal to highly exceptional. Most people are more or less average with respect to any given aptitude. A few of them are considered to have very low aptitudes, and another small group manifests outstanding aptitudes in one or more domains. As expected, I reserved the label gifted for those who occupy the top end of the aptitude continuum. Because of the multiple domains, giftedness has many faces: people can be intellectually gifted, creatively gifted, socially gifted, and so forth. Without surprise, I also fixed the minimum threshold at the top 10 % on any valid measure of aptitude. And within that top group, I propose similar levels of giftedness (see Figure 2): top 1 % (highly gifted), top 0,1 % (exceptionally gifted), and top 1:10,000 (extremely gifted).

Intellectual aptitudes. The theme of this chapter is academic talent development (ATD). In connection with ATD, one aptitude domain stands out: the intellectual domain. But what is intelligence? There are so many definitions proposed and so many types of intelligences, for instance Gardner's (1983) multiple intelligences, Goleman's (1995) emotional intelligence, Sternberg's (2005) successful intelligence, and many others. Thankfully, a group of 52 eminent cognitive psychologists have proposed a clear consensual answer to that question. In December 1994, in a full page of the Wall Street Journal, they published a position statement on the nature of intelligence (Gottfredson, 1997). It was titled Mainstream Science On Intelligence (MSOI). It comprises 25 short articles, each of them focusing on a

specific question, for instance the definition and measure of intelligence, individual and group differences, the impact of intelligence in education, in the work place, and in daily life. The first article specifies what they mean by intelligence; it is by far the definition I prefer.

> "1. Intelligence is a very general mental capability that, among other things, involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly and learn from experience. It is not merely book learning, a narrow academic skill, or test-taking smarts. Rather, it reflects a broader and deeper capability for comprehending our surroundings – 'catching on,' 'making sense' of things, or 'figuring out' what to do." (p. 13)

That definition identifies seven specific and very concrete thinking abilities. Among them, learning easily, thus quickly, is in my view the key behavioral trademark of high intelligence. In their second article, the authors of the MSOI make an important precision.

> "2. Intelligence, so defined, can be measured, and intelligence tests measure it well. They are among the most accurate (in technical terms, reliable and valid) of all psychological tests and assessments. They do not measure creativity, character, personality, or other important differences among individuals, nor are they intended to." (p. 13)

This second article is very important because it clarifies the conceptual limits of that concept, which characteristics are NOT components of their definition. For instance, intelligent individuals do not necessarily possess high creativity, personal maturity, or judgment and wisdom.

Why is intelligence automatically associated with school achievement? There is a very simple reason. Since the creation of the first IQ test over a century ago, thousands of studies (Jensen, 1998; Macintosh, 2011) have shown that individual differences in cognitive abilities predict academic achievement much better than any other causal influence, except of course past or current academic achievement. For example, correlations between IQ scores and elementary school academic results range between .60 and .70; they decrease at the high school level, but still range between .50 and .60. These correlations are just one typical example of the strong relationships that link specific aptitudes with specific achievements; in every learning situation, aptitudes act as the building materials for specific types of competencies. For instance, muscular aptitudes are the building blocks of sporting skills; high creative aptitudes will differentiate those who excel in visual arts or music composing from those who do not; high social aptitudes will improve the competencies of people whose work brings them in constant interaction with other people (e.g., teachers, nurses, salespeople, politicians). In fact, we could define any type of learning process as the progressive transformation of some aptitudes into related competencies.

1.3 Developmental dynamics

How does this process happen, and which are the main actors involved? The DMGT figure illustrates that causal relationship by linking aptitudes to occupational fields with a developmental process that comprises three elements: a) learning activities (e.g., curriculum, pedagogical strategies, environmental conditions); b) investments in time, energy, and money; c) indices of progress (e.g., stages, learning pace, and occasional favourable or unfavourable turning points). These three basic components, the aptitudes (gifts), the competencies (talents), and the developmental process, are not alone: two other components (see Figure 1) contribute substantially to the progressive construction of competencies and talents. They are called catalysts because they are not building blocks of competencies; their role is to help (or sometimes block) the developmental process.

The first group of catalysts, called intrapersonal, brings together the learners' personal characteristics; they cover a huge diversity. At the top (see Figure 1) appear two groups of traits: physical traits (e.g., handicaps, general health) and mental traits (temperament and personality). Personality traits that have been studied as potential predictors of school achievement are: anxiety, passive or explicit aggressiveness, perfectionism, conscientiousness or dependability, intellectual curiosity, and many others. Most of these studies have obtained inconsistent or modest results (Neihart, Pfeiffer, & Cross, 2016). Three constitutive components of the goal management process appear underneath: a) awareness, especially of one's strengths and weaknesses; b) motivation, defined as goal identification through a choice of personal motives (e.g., values, needs, interests and passions); c) and volition or will power, defined as goal attainment (e.g., perseverance, determination, grit, resilience). Just like in the case of aptitudes, countless studies have confirmed that all these personal characteristics have substantial hereditary origins (McCrae, 2009).

The second set of catalysts brings together all relevant environmental influences. The DMGT subdivides them into three main dimensions: a) the milieu or socioeconomic dimension (e.g., social, political, financial, and cultural influences; the family's socio-economic status); b) significant individuals or the psychological dimension (e.g., parents, siblings, teachers, peers, mentors); c) and the educational dimension in the form of resources (e.g., curriculum, ability grouping, special schools, targeted services, special facilities like libraries, computers, or sports centers).

You have certainly noticed a large background area titled 'chance.' Its inclusion means that individuals have only partial control on the various factors that will affect their talent development. As a famous psychologist once said: "All human accomplishments can be ascribed to 'two crucial rolls of the dice over which no individual exerts any personal control. These are the accidents of birth and background. One roll of the dice determines an individual's heredity; the other, his formative [family] environment' (Atkinson, 1978, p. 221)." These two impacts alone

give a powerful role to chance in favouring or hindering a person's talent development possibilities.

The growth of competencies in a particular field of occupation or in academic subjects will depend on very complex dynamic interactions between dozens of influences belonging to these four causal sources: aptitudes, developmental process, intrapersonal, and environmental catalysts. Each person follows a personal and unique developmental path, marked by constant personal or environmental influences; they will affect the probability of reaching talent-level achievements. Keep in mind that competencies have multiple causal sources. Consequently, someone may reach talent-level achievements (top 10 %) without having gifted-level aptitudes (top 10 %). Individuals can compensate these above average, but not remarkable, aptitudes with strong support from both intrapersonal and environmental catalysts. In others words, many high achievers, but not all of them, are gifted; on the other hand, all high achievers, those in the top 10 %, are by definition talented. Unfortunately, time constraints do not allow a more detailed discussion of the complex dynamic relationships between the DMGT's five components. Interested readers will find a more detailed discussion of the DMGT in almost every handbook about gifted education, but especially in the forthcoming DMGT 'bible' (Gagné, 2020).

2. From theory to practice

Concerning the practical implications of the DMGT, I will first illustrate the longitudinal impact of individual differences in ease/speed in learning when, of course, they are not artificially controlled by the school system.

2.1 The breadth of academic competencies

You probably studied in high school a basic law from the physics of motion: Distance is the product of Speed x Time. In other words, the faster you go the farther you should be as the length of time increases. Think of marathon runners; they all begin the race tightly grouped at the starting line. As the race progresses, the distance increases between the fastest and the slowest runners. Currently, an international marathon is won in just over two hours, whereas the slowest amateurs will take easily three times as much to complete it. It is the same with all learning situations: the fastest learners (a.k.a. talented students) should master, as time goes by, more advanced competencies. Please note that I did not substitute speed with aptitudes; as stated in the DMGT, the fastest learners are those who benefit not only from high aptitudes, but also from the strong support of intrapersonal and environmental catalysts, and also from a strong investment in their developmental

process. I insist: there is more to high academic achievement than just intellectual giftedness!

Figure 3 illustrates this phenomenon; it is called the 'fan spread' effect. Years of schooling are at the bottom, and academic progress (in school grades) is on the vertical. The fan-spread effect is just the normal expression of differences in learning pace. In the figure, the diagonal represents average learners, the area below the diagonal slow learners, and the area above fast learners. The slowest learners in that graph have mastered only half of the expected curriculum, namely the curriculum of 4 grades in 8 years of schooling; on the other hand, the fastest learners have mastered twice the expected curriculum, namely the equivalent of 8 grades in 4 years. The best learners are progressing four times faster than the slowest ones. We can see the vertical gap in achievement grow from one year to the next. Of course, that increasing gap depends on one important condition: all students are allowed to progress at their own learning pace. It is so common in learning situations that scholars call the fan-spread effect the first law of learning (Gagné, 2005; Jensen, 1991).

The slowest learners are easy to identify; every teacher knows students who, after the normal six years of elementary schooling, are still functionally illiterate or lack basic numeracy knowledge and skills. But, how about learning twice as fast, for instance reaching Grade 8 knowledge mastery in just four years of learning? Isn't that totally utopic? Not at all. Just think of young students who enter Grade 1 and already know how to read and do basic arithmetic; many even already achieve as well as an average Grade 3 student. That is an example of twice as fast. But, can they keep up that pace over many years? The fastest learners could do it, if we left them free to do so; and sometimes, unfortunately too rarely, the system allows them to do so.

Here is a personal example. I am currently following a young girl in Montreal, named Catherine. Table 1 shows her academic progress. Born in December 2006, she was 6,5 years old in June 2013, and should have just completed Kindergarten; but, she entered Kindergarten a year early, and then completed very easily the first two grades in her first year of primary schooling; she was ready to move up to Grade 3. She continued to successfully complete two grades each year; so, in June 2015, she was ending Grade 6, four years ahead of her age peers. She continued her 'two in one' learning pace without any difficulty, completing Grade 10 in June 2017. Her age should have placed her in Grade 4, like the table shows; in fact she was six years advanced, as well as six years younger than her classroom peers. In the fall of 2017, still 10 years old, she began a college course while completing high school; she then entered college full-time in January 2018, which is Grade 12 in Quebec, at the beginning of age eleven! Thanks to her parents' determination and the openness of a few local educators, Catherine has been able to progress at the learning pace she wanted: twice as fast as most students. She holds the equivalent of a Guinness record in Quebec! And, by the way, she is absolutely happy and

socially adapted, thanks in large part to that opportunity she was given to progress at her desired learning pace.

The Talent Searches in the United States offer another fascinating example of self-acquired, highly advanced mastery of the regular curriculum (Lupkowski-Shoplik, Benbow, Assouline, & Brody, 2003). Each year, high achieving students in Grades 7 and 8 can test their level of advanced curriculum mastery. They are invited to pass the SAT (or the ACT), a test that all high school seniors must complete as part of their admission process to US colleges. There are two versions, one for mathematics (SAT-M) and one for language (SAT-V). With its focus on reasoning more than memorized knowledge, the SAT can be considered halfway between an aptitude test and an achievement test. These talented students are 4 or 5 years younger than high school seniors, and have not formally studied the math or language curriculum that is supposed to prepare them for that difficult exam.

The originators of this competition were convinced that very talented students faced a ceiling effect in their regular exams: there were not enough difficult questions to test the limits of their high level mastery. As shown in Figure 4, they were hoping that this high-level competition would recreate within that group of almost identically talented students a new normal curve of achievements. Their hopes were fully confirmed. Year after year (see Table 2), about 25 % of these talented Grade 7 students and almost 50 % of the talented Grade 8 students obtain SAT scores that are equal or exceed those of average high school seniors who are continuing to college. Many of them even obtain SAT scores high enough to gain admission in the best universities. Remember that these exceptionally high achieving students are not only 4 or 5 years younger than high schools seniors, but they have not studied (officially of course!) the high school curriculum that would prepare them for that difficult test.

Here is a last example. Some years ago I analysed the norms for a very popular achievement test called the Iowa Tests of Basis Skills (ITBS) (Gagné, 2005). They offer achievement batteries that cover the basic curriculum from Grades 1 to 9. The raw scores on any of these tests can be transformed into a single scale of scores called 'developmental;' that special scale makes it possible to compare the academic competencies of all students independent of their grade level. Figure 5 shows the score distribution of a representative US sample of 20,000 Grade 5 students on that developmental scale; the samples for all other grade levels were about the same size. This figure shows that one third of Grade 5 students in the USA obtained on the ITBS achievement results at least equal to those of average Grade 6 students, that 20 % of them achieved at the level of average Grade 7 students, that 10 % were advanced by 3 academic years, and that 4 % of these 5fth graders – one per classroom on average! – could compete academically with average Grade 9 students. In other words, if we allowed students to progress at the pace of their aptitudes, motivation, and effort, talented learners could skip many grades without any difficulty in terms of subject mastery.

2.2 The constraints imposed by the K-12 school system

Does the educational system take into consideration the breadth of student competencies, their fan-spread growth over the school years, and especially the advanced mastery of the fastest learners? Absolutely not! First, the school system forces all students to follow the pace of what the famous psychologist Julian Stanley called an 'age/grade lockstep' (Stanley, 1979); if you are 8 years old, you are in Grade 3, if you are 11 years old you are in Grade 6, and so forth. Each year you move ahead one grade level whatever your academic achievement results. The initiators of this age/grade lockstep never noticed Aristotle's admonition, subsequently repeated by Thomas Jefferson: "There is nothing more unequal than the equal treatment of unequal people."

The only students who do not follow this age/grade lockstep are those whose very slow pace forces them to repeat a grade. In the Quebec elementary school system, as shown in Table 3, approximately 93 % of all students follow the age/grade lockstep, 98 % in Grade 1, 92 % in Grade 3, and 88 % in Grade 6, the last year of elementary school. Another 6,5 % are below their expected grade level, growing from 1 % in Grade 1 to about 11 % in Grade 6; you can observe a clear fan-spread effect, but only among the slow learners! Only 1,5 % of these 235,000 students have advanced beyond their normal grade level, in most cases thanks to a local program of early entrance to Kindergarten. In other words, grade skipping is an exceedingly rare commodity in Quebec! But I am convinced we would find similar statistics in all school districts around the world, including of course Germany!

Second, many studies have shown (see Reis et al., 1993) that in most classrooms, the teachers follow the learning pace of below average students, those around the 25th or 30th percentile. This slow pace aims to increase the percentage of students who will 'succeed' at the end of the year, and will thus remain within the age/grade lockstep. But think of the large percentage of students, not just that top 10 % of talented students but many more, who suffer, day after day, continuous repetitions of lessons they have already well mastered. They never get a chance to confront new challenges, as well as occasional failure; what they risk learning instead is laziness and mediocrity. Third, because most teachers focus their attention on slow learners and special education students, they rarely think of offering their talented students appropriate enrichment. Major evaluation studies [e.g., Archambault et al., 1993.) have confirmed the rarity of such enrichment. These students will have to satisfy themselves with occasional pull-out classes, weekend activities, or summer camps. But, the bulk of their school week, spent in the regular age/grade classroom, will give them few occasions to nourish their interests and advanced learning abilities.

Teachers usually alleviate any guilt feelings by pointing out that their talented students, being so bright, will succeed anyway! Of course, most of them will succeed, although a small percentage of these gifted learners will become so bored and frustrated that they will start underachieving. My main point is that those

who continue to achieve will progress much more slowly than what their fast learning pace would make possible. In other words, they do not benefit from resources that would foster the development of their talents to their fullest potential. Fortunately, the examples I presented earlier indicate that some of them find ways, completely outside the regular school curriculum, to progress at an accelerated pace and demonstrate the potentialities of a full-blown fan-spread distribution. But that remains a possibility, not a reality!

2.3 Toward 'real' academic talent development programs

Let me now describe what we could do to develop to their fullest potential the intellectual gifts and intense interests of these young talented learners, as well as those of less exceptional school learners. What I will describe are general guidelines; they leave many administrative details to be identified and worked out at the local level. Such a system would have as its conceptual basis a profound respect for individual differences in pace of learning, as they manifest themselves through the fan-spread effect. Because of that, I fully endorse the educational ideology proposed by the late Elliot Eisner (1933–2014), a renowned former professor of education at Stanford University. Here is a short excerpt of his ideas on appropriate learning conditions for all students, but especially for the usually overlooked fast-learning talented students.

> "The kind of schools we need would not hold as an ideal that all students get to the same destinations at the same time. (…) Individuals come into the world with different aptitudes, and, over the course of their lives, they develop different interests and proclivities. In an ideal approach to educational practice, (…) each youngster would learn at an ideal rate. (…) Over time, the cumulative gap between students would grow. Students would travel at their own optimal rates, and some would go faster than others in different areas of work." (Eisner, 2002, p. 580)

We have in this short excerpt a clear description of a fully functioning fan-spread effect. Now, if we focus on our special group of quick learners, how would we structure their academic activities to help them develop "their talents and mental and physical abilities to their fullest potential," as stated in the United Nations' Convention of the Rights of the Child? Here are seven constitutive components of what I would call a 'real' Academic Talent Development program.

1. Enrichment of the regular classroom curriculum;
2. Daily implementation of that enrichment;
3. Full-time ability grouping;
4. Personalized (accelerated) pacing;
5. Long-term excellence goals;
6. Selective access to the program;

7. Early implementation.
Let's look at them more closely.

Curricular enrichment. First and foremost, academic talent development programs aim to foster academic excellence, and academic excellence expresses itself as outstanding mastery of the K–12 curriculum. Consequently, it is that basic curriculum we must enrich for academically talented students to experience regular learning challenges. Provisions that do not implement this keystone characteristic cannot receive the ATD label. What does an enriched curriculum look like? I have proposed (Gagné, 2007) four different types of enrichment: enrichment in Density, in Difficulty, in Depth, and in Diversity. That particular sequence reflects a decreasing order of relevance, thus giving clear priority to enrichment in Density. Also called curriculum condensation or curriculum compacting, it consists in accelerating the pace of content coverage in all subject matters. Academic talent development specialists should give it priority over other forms of enrichment because it offers the most relevant response to giftedness' trademark, namely ease/speed in learning. Moreover, the school time liberated through faster mastery of subject matter units creates learning space for additional enrichment (Renzulli, Smith, & Reis, 1982).

Daily enrichment. Talented students manifest their fast learning pace every day and all day, not episodically. Consequently, appropriate enrichment must propose intellectual challenges on a daily basis. The concept of zone of proximal development advanced by Lev Vygotsky (1978) aptly conveys the need to maintain the students' learning pace at the cutting edge of their learning capacity. Teachers must watch constantly for signs of unchallenging content; if there is one thing that most high-achieving motivated students resent, it is to face, day after day in the regular classroom, the slow and repetitious tempo imposed by slow learning peers. This focus on daily enrichment excludes from ATD programs most popular so-called 'gifted programs' that are offered on a part-time basis, like weekly pull-out classes, weekend enrichment activities, or summer camps. It does not reduce in any way their value as useful additions to ATD programs for motivated students who want to pursue their learning beyond the regular classroom, but they cannot be considered appropriate responses to the constant enrichment needs of talented students.

Full-time grouping. How can we best deliver daily enrichment to these talented students? The most effective and efficient solution is to group them with a specially trained ATD teacher. Unfortunately, this very sensible administrative solution remains very controversial. On the other hand, there is ample research evidence on the positive academic impacts of ability grouping, as long as it serves to offer proper enrichment to these students (Kulik, 2003). At the same time, parallel research, both in the USA and Australia (Archambault et al., 1993; Whitton, 1997), has demonstrated the almost total lack, in regular classrooms, of enrichment activities that specifically target academically talented students. These eval-

uation studies have shown that the vast majority of these provisions offer little more than a lip service response to talented students' learning needs. The results revealed, among other things, that teachers offered these activities no more than two or three times a month. Even worse, the activities usually targeted the whole classroom, leaving little specific enrichment for talented students. Teachers felt much more pressure to attend to the learning difficulties of slower learners and of students with special learning or behavioral difficulties; they did not consider the needs of their high achieving students to be a priority.

Personalized pacing. Grouping students to offer an enriched curriculum does not mean that individual differences in learning pace have disappeared; remaining differences produce over time an increasing gap between the slower and faster talented learners. Consequently, those who progress significantly faster should be allowed to move ahead at an accelerated pace, if they so desire of course. Unfortunately, most accelerative measures face strong resistance from a majority of administrators, teachers, and parents; they ignore or refuse to accept the overwhelming scientific evidence in support of all forms of accelerative enrichment. Borland (1989) elegantly summarized that conundrum when he affirmed: "Acceleration is one of the most curious phenomena in the field of education. I can think of no other issue in which there is such a gulf between what research has revealed and what most practitioners believe" (p. 185).

Long-term excellence goals. Excellence goals must be defined with respect to the enriched school curriculum; they should also be expressed normatively, which means in relationship with the expected achievements of learning peers. Their normative status distinguishes these goals from the attainment of "personal bests" that can apply to the academic goals of all students. These long-term educational goals that talented students would choose themselves should far exceed the level of academic excellence typically expected within the regular curriculum. Obtaining high marks in a regular classroom has nothing to do with academic talent development; most academically talented students can reach such goals without any effort.

Selective access. Academic talent development requires not only outstanding natural learning abilities, but also demonstrated probability of future success. Unless the selection ratio is placed at a high level, for instance my proposed threshold of top 10 %, many selected students could fail to perform at the level expected by the enriched curriculum. It is then tempting to reduce the program's performance requirements to avoid forcing these students to leave. Program administrators should resist that temptation because it marks the beginning of a slippery course toward reduced expectations. Instead, all selected students should know before being admitted into the ATD program that their continued participation depends on maintaining adequate academic results. On the other hand, any selection process leaves ample room for error; this suggests the initial admission of a larger percentage of students, reminding them that it will be their responsibility to keep pace with all others.

Early implementation. A fundamental law of individual differences in development states that precocity manifests itself … precociously! Many children who enter kindergarten already know the alphabet, can write their name, read some words, and even do simple arithmetic computations. Their intellectual precocity makes them better prepared than average Grade 1 students to tackle the regular school curriculum. Consequently, this seventh constitutive element strongly invites school administrators to adopt as the cornerstone of their school district's ATD program a policy allowing early entrance to kindergarten or primary school. Of course, qualifying early entrance as a "cornerstone" implies that it will be followed by the other building blocks of a comprehensive ATD pathway, all the way from kindergarten to college.

3. Conclusion

I began this chapter by proposing a conceptual framework, the DMGT, which aims to foster a better understanding of the complexities of talent development, a process that consists essentially in transforming high aptitudes into high competencies. It should also be understood as a map of all causal influences contributing to these remarkable outcomes that are talents in every occupational sphere. I then focused on academic talent, and demonstrated that differences in learning facility lead automatically, as times progresses, to an increasing gap between the slowest and fastest learners. I tried to illustrate through numerous examples, from little Catherine to Talent Searches, how huge this gap can become. It can happen even within an educational system that implements for all students a strict age/grade lockstep curriculum, ignores appropriate enrichment in density, and puts major constraints on the free expression of individual differences in learning pace. Within any grade level, the competencies of that top 10 % labelled talented can cover the curriculum of many grade levels. I finally proposed a series of best practices founded on the fan-spread ideology, which means a profound respect for the maximal expression of individual differences in learning ability. I strongly believe that these seven best practices would directly contribute to the maximum fulfilment of those talented students who harbour high intellectual aptitudes, strong interests and well-defined passions. Note also that most of these best practices, especially ability grouping and accelerative enrichment, add little cost to the educational system.

At the same time, I am well aware that these goals might appear to some administrators rather 'elitist' and far from more urgent educational priorities. Yet, it would be worthwhile to establish as soon as possible, and at the highest levels of all educational systems, clear long-term goals for academic talent development. And these goals should be based on clear conceptual views and definitions of the target population. First, it would affirm that academically talented students, from kindergarten to high school, represent a target population worthy of special at-

tention, thus sending a clear message to regional and local educational structures. Second, it would show which main directions the educational authorities intend to follow in their curricular and administrative responses to these students' special educational needs. I hope that these theoretical and practical suggestions will help all talented students, in the short and long-term, maximize their potentialities to their fullest.

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Table 1: Catherine's educational progress

Date	Age	Normal grade level	Catherine	Advance
06-2013	6,5	Preschool	Grade 2	+ 2
06-2015	8,5	Grade 2	Grade 6	+ 4
06-2017	10,5	Grade 4	Grade 10	+ 6
06-2018	11,5	Grade 5	1st semester in college	+ 6,5

Table 2: Some results from Talent Searches

NORMS	SAT-V	SAT-M
Average for *male* college-bound high school seniors	507	533
Average for *female* college-bound high school seniors	504	498
How many (%) equaled or exceeded these averages?		
7th grade *male* participants	22	27
8th grade *male* participants	47	50
7th grade *female* participants	24	32
8th grade *female* participants	47	59

Data from: Lupkowski-Shoplik, Benbow, Assouline, & Brody, 2003, p. 207.

Table 3: Student distribution in Quebec's education system (2008)

Age at 09/30	Grade levels (K = thousands)								
	P	1	2	3	4	5	6	7	8
6	**779**	**72 K 98%**	**825**	**3**	**2**				
8		**159**	**4805**	**70 K 92%**	**1012**	**14**	**5**		
11		**46**	**432**	**276**	**1158**	**6847**	**76 K 88%**	**1662**	**68**

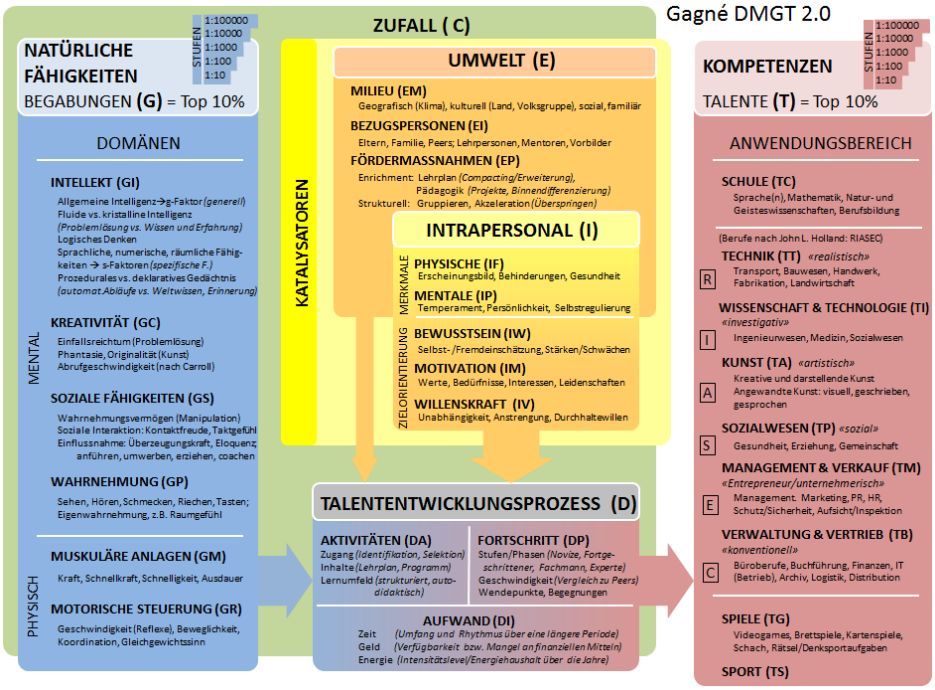

Figure 1: Differenzierendes Modell von Begabung und Talent.

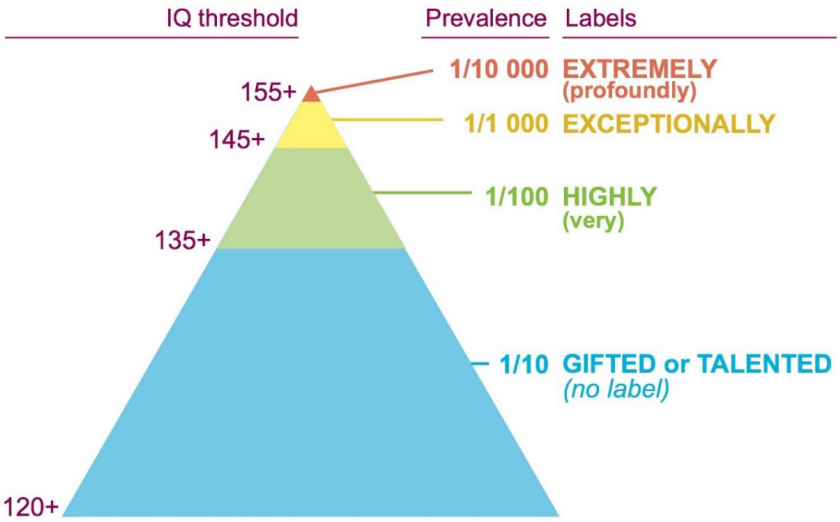

Figure 2: 4-level MB system (gifts and talents).

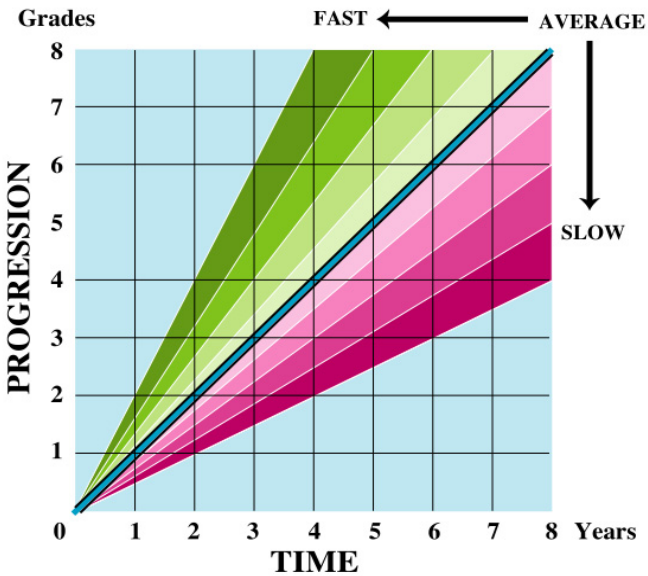

Figure 3: The fan-spread effect.

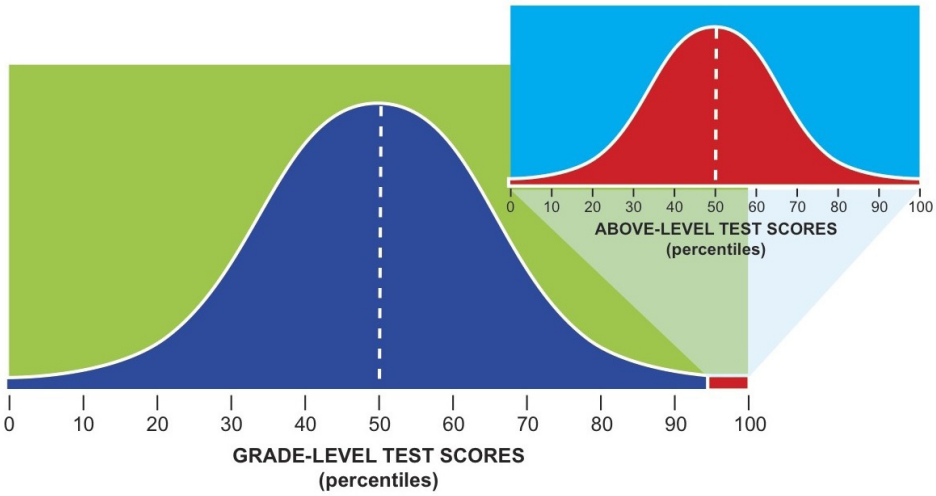

Figure 4: Breaking the ceiling effect.

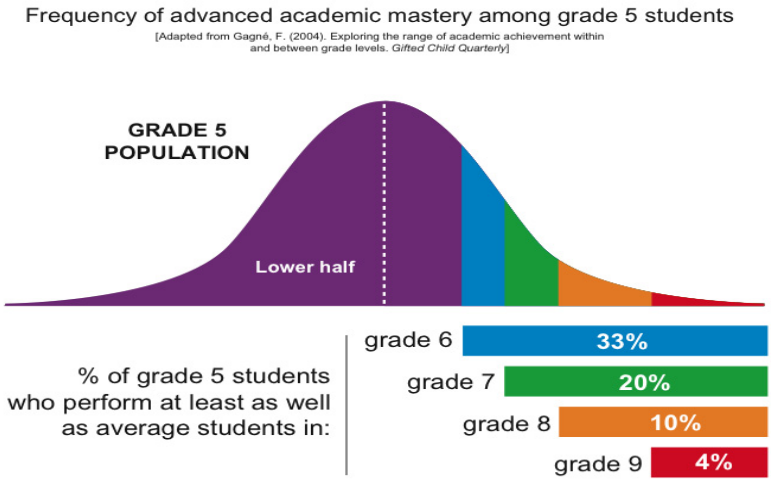

Figure 5: Frequency of advanced academic mastery among Grade 5 students.

Data adapted from Gagné, 2005, p. 145.